

Full Worked Solutions — All Three Assignments

This document walks through **every single question** across all three uploaded assignments — Assignment 1 (Visualization & Central Tendency), Assignment 2 (Probability, Unit 4), and Assignment 3 (Bayes' Theorem, Unit 8) — with complete, step-by-step reasoning. Nothing is skipped; every intermediate number is shown so you can trace exactly how each final answer was reached. Formulas are in LaTeX (Obsidian-compatible).

ASSIGNMENT 1 — Unit 1 (Visualization) & Unit 2 (Central Tendency)

Question 1 — Line Diagram of Accident Data

Given:

Number of accidents	0	1	2	3	4	5	6	7	8	9	10
Number of drivers	82	44	68	41	25	20	13	7	5	4	3

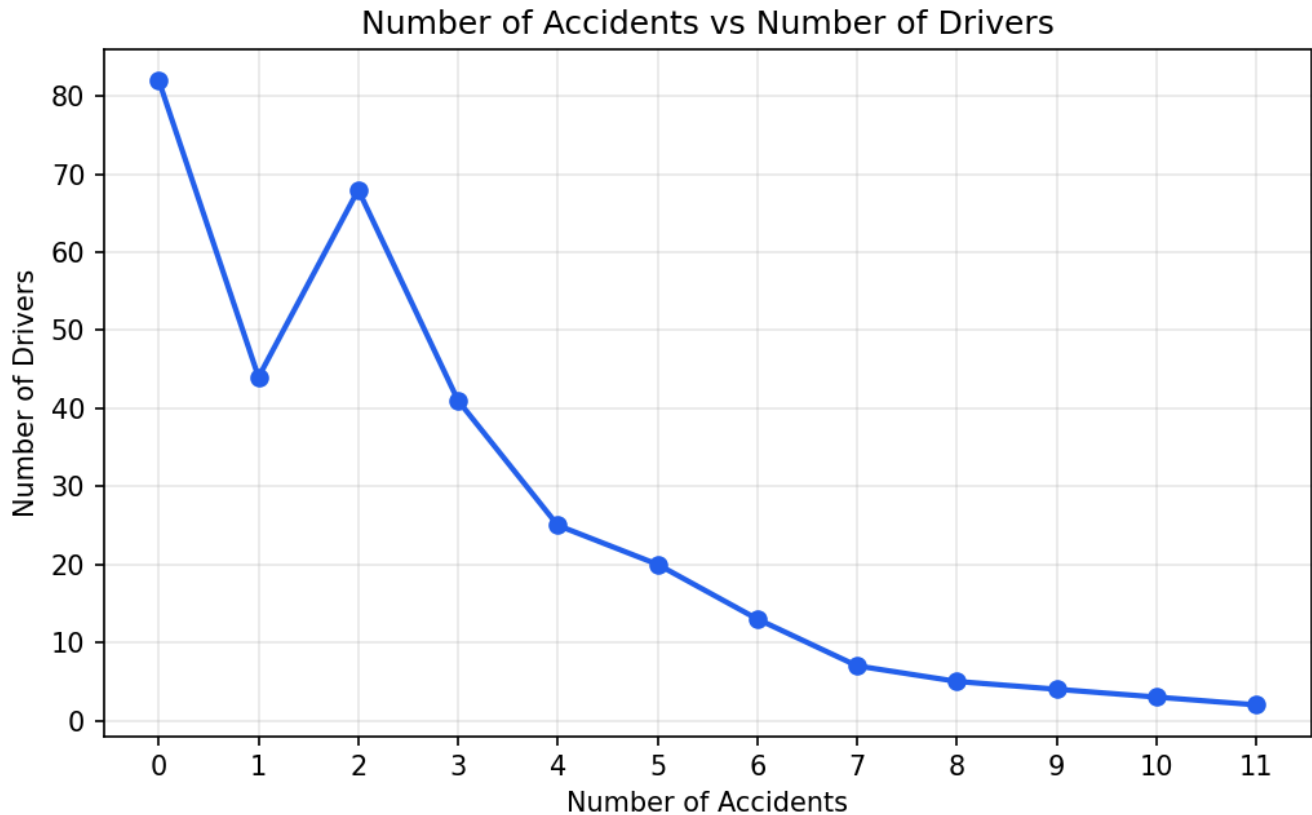
How to think about this problem: A line diagram is the right tool here because "number of accidents" is a **discrete** variable (you can't have 2.5 accidents), and we want to visually communicate how frequency (number of drivers) *changes* as the number of accidents increases. Unlike a histogram — which is reserved for continuous, grouped data where bars must touch to represent an unbroken range — a line diagram simply places a point at each discrete x-value and connects consecutive points with straight line segments. This makes the *trend* (rather than the *area*) the visually dominant feature, which is appropriate since there's no "in-between" value of accidents to worry about representing.

Step-by-step construction:

1. Draw the horizontal (x) axis and label it "Number of Accidents," marking values 0 through 11 at equal spacing.
2. Draw the vertical (y) axis and label it "Number of Drivers." Since the values range from 2 up to 82, choose a scale (e.g., increments of 10) that comfortably fits all points.

3. Plot a point at each (accidents, drivers) pair: (0,82), (1,44), (2,68), (3,41), (4,25), (5,20), (6,13), (7,7), (8,5), (9,4), (10,3), (11,2).
4. Connect each consecutive pair of points with a straight line, in order of increasing x.

Resulting diagram:



Reading the diagram: The line rises sharply from 0 to 1 accident is actually a *dip* first (82 → 44), then spikes back up at 2 accidents (68) before falling in a long, steady, decaying tail out to 11 accidents. This shape tells a real story: most drivers had either zero accidents or a moderate number (around 2), and very few drivers had many accidents — the distribution is heavily right-skewed (a small number of "high-accident" drivers pull a long tail to the right), which is typical of accident/claim-count data in insurance and safety statistics.

Question 2 — Histogram

Given:

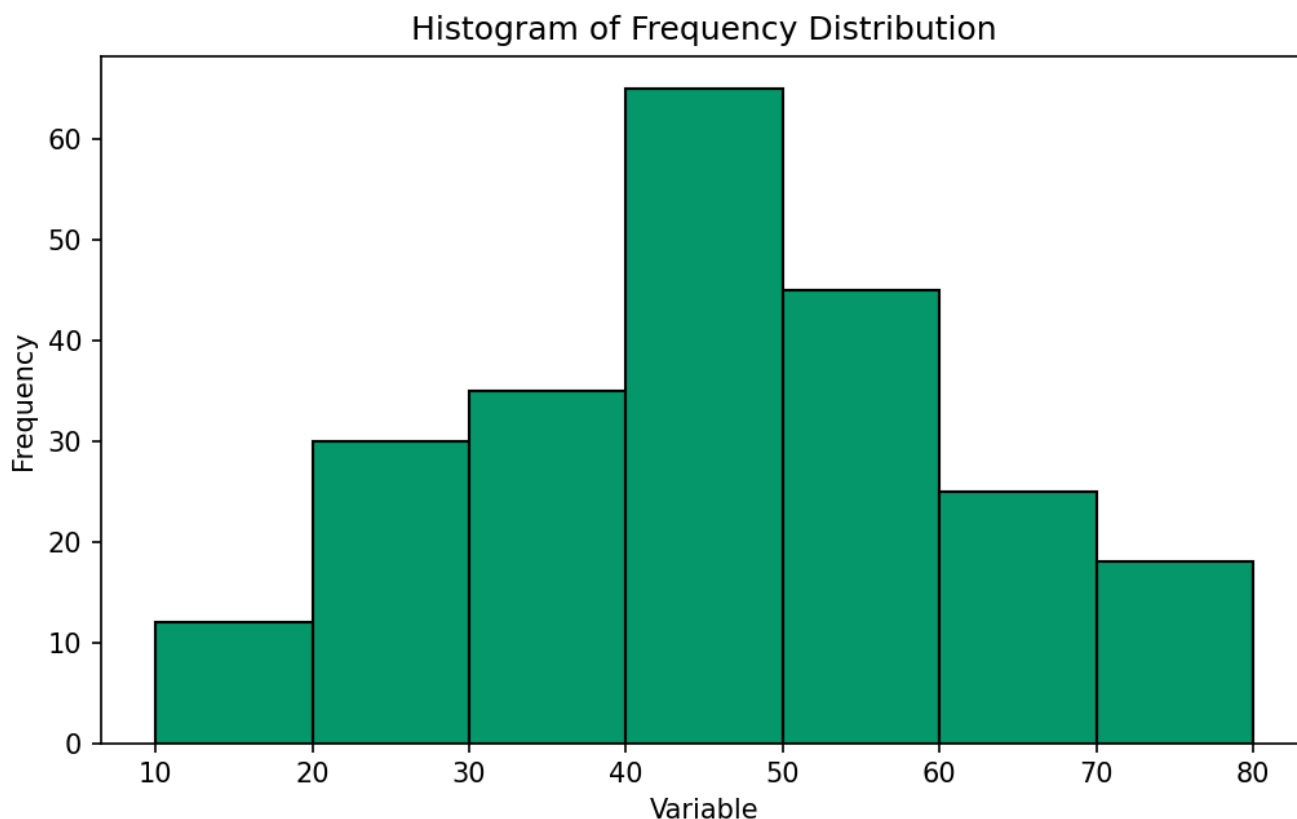
Variable	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	12	30	35	65	45	25	18

How to think about this problem: Unlike Question 1, this data is **continuous and grouped into class intervals** (10–20, 20–30, etc.), so a histogram — not a line diagram — is the correct chart. In a histogram, bars sit directly against each other with no gaps, because the classes represent a continuous, unbroken range of the underlying variable; there's no "missing" value between, say, 19.99 and 20.00. All the class widths here are equal (each is 10 units wide), so the bar heights can be set directly equal to the frequencies without needing to compute frequency density.

Step-by-step construction:

1. Draw the x-axis and mark the class boundaries: 10, 20, 30, 40, 50, 60, 70, 80.
2. Draw the y-axis and label it "Frequency," with a scale that comfortably spans 0 to 65 (the largest frequency).
3. For each class interval, draw a bar whose left edge sits at the class's lower boundary and whose right edge sits at the class's upper boundary (so each bar is 10 units wide), with height equal to that class's frequency:
 - 10–20 → height 12
 - 20–30 → height 30
 - 30–40 → height 35
 - 40–50 → height 65
 - 50–60 → height 45
 - 60–70 → height 25
 - 70–80 → height 18
4. Push all bars flush against their neighbors (no gaps between bars) — this is what visually distinguishes a histogram from a bar chart and signals that the underlying variable is continuous.

Resulting diagram:



Reading the diagram: The distribution rises from a modest 12 in the 10–20 class up to a clear peak of 65 in the 40–50 class, then falls away again down to 18 in the 70–80 class. This gives a roughly bell-shaped (though not perfectly symmetric) distribution, with the bulk of the data concentrated in the middle classes (30–60) and progressively less mass in the tails.

Question 3 — Pie Diagram of Government Expenditure

Given (in million Rs.):

Item	Agriculture & Rural Development	Industries & Urban Development	Health & Education	Miscellaneous
Proposed Expenditure	4,200	1,500	1,000	500

How to think about this problem: A pie chart is the right tool whenever the goal is to show how a **whole quantity breaks down into parts** — here, the whole is total proposed government expenditure, and we want to visually communicate what *share* each category commands. Every slice's angle is proportional to that category's share of the 360° full circle.

Step 1 — find the total:

$$\text{Total} = 4200 + 1500 + 1000 + 500 = 7200 \text{ million Rs.}$$

Step 2 — find each category's proportion of the total, then convert that proportion into a sector angle using:

$$\text{Sector angle} = \frac{\text{value of category}}{\text{total}} \times 360^\circ$$

Doing this for each category:

Agriculture & Rural Development:

$$\frac{4200}{7200} \times 360^\circ = 0.58333 \times 360^\circ = 210.0^\circ$$

(This is 58.33% of the total budget.)

Industries & Urban Development:

$$\frac{1500}{7200} \times 360^\circ = 0.20833 \times 360^\circ = 75.0^\circ$$

(This is 20.83% of the total budget.)

Health & Education:

$$\frac{1000}{7200} \times 360^\circ = 0.13889 \times 360^\circ = 50.0^\circ$$

(This is 13.89% of the total budget.)

Miscellaneous:

$$\frac{500}{7200} \times 360^\circ = 0.06944 \times 360^\circ = 25.0^\circ$$

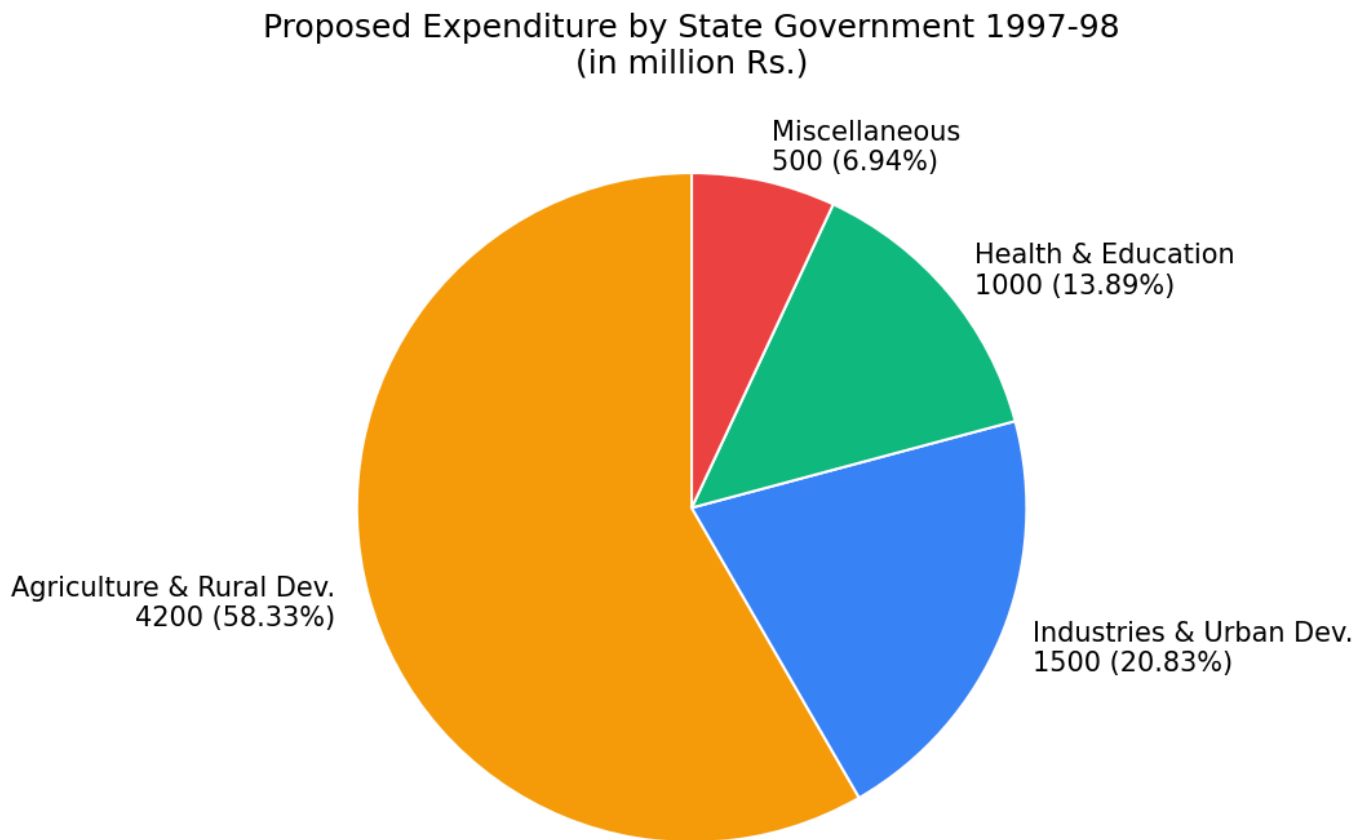
(This is 6.94% of the total budget.)

Step 3 — sanity check: all sector angles must sum to exactly 360° :

$$210.0^\circ + 75.0^\circ + 50.0^\circ + 25.0^\circ = 360.0^\circ \checkmark$$

Step 4 — construct the diagram: draw a circle, pick a starting radius (conventionally at the 12 o'clock position), and mark off each sector's angle in sequence, moving clockwise, using the angles computed above.

Resulting diagram:



Interpretation: Agriculture & Rural Development dominates the budget at just over half (58.3%), more than double the next-largest category (Industries & Urban Development at 20.8%). Health & Education and Miscellaneous together make up less than a fifth of the total proposed expenditure.

Question 4 — Weighted Mean of Sales Growth Rates

Given:

Branch	Current Sales (millions)	Expected Growth
Atlantic	\$193.8	7.25%
Midwest	\$79.3	8.20%
Pacific	\$57.5	7.15%

Why a simple average is wrong here: If you simply averaged 7.25%, 8.20%, and 7.15%, you'd get $(7.25 + 8.20 + 7.15)/3 = 7.5333$ — but this treats all three branches as equally important, when in reality the Atlantic branch (at \$193.8M) is more than three times the size of the Pacific branch (at \$57.5M). A branch's growth rate should influence the *company-wide* average growth rate in proportion to how much of the company's total sales that branch represents — that is exactly what a **weighted mean** does, using each branch's current sales figure as its weight.

Formula:

$$\bar{x}_w = \frac{w_1x_1 + w_2x_2 + w_3x_3}{w_1 + w_2 + w_3}$$

where each x_i is a branch's growth rate and each w_i is that branch's current sales (the "weight").

Step 1 — compute each weight × value term (numerator components):

$$193.8 \times 7.25 = 1405.05$$

$$79.3 \times 8.20 = 650.26$$

$$57.5 \times 7.15 = 411.125$$

Step 2 — sum these products:

$$1405.05 + 650.26 + 411.125 = 2466.435$$

Step 3 — sum the weights (total current company-wide sales):

$$193.8 + 79.3 + 57.5 = 330.6$$

Step 4 — divide:

$$\bar{x}_w = \frac{2466.435}{330.6} = 7.4603$$

Final Answer: 7.46% — this matches the given answer. Note that this weighted result (7.46%) sits closer to the Atlantic branch's growth rate (7.25%) than the simple average (7.53%) would suggest, precisely because the Atlantic branch carries the most weight in the company's overall sales.

Question 5 — Geometric Mean of Bad Debt Growth

Given:

Year	1989	1990	1991	1992	1993	1994	1995
Value	0.11	0.09	0.075	0.08	0.095	0.108	0.120

How to interpret the table: Each value in this row is itself an **annual growth rate** (in decimal form) for bad debt expense in that year — e.g., in 1989, bad debt expense grew by 11%; in 1990, it grew by 9%; and so on through 1995's growth of 12%. Because these are seven *separate yearly growth rates* (not seven raw expense levels), the right way to find a single "average annual growth rate" that summarizes all seven years is the **geometric mean of the growth factors**, not an arithmetic average of the raw percentages — averaging percentages arithmetically would not correctly reflect how growth *compounds* year over year.

Step 1 — convert each yearly growth rate into a growth factor by adding 1 (a growth rate of r corresponds to a growth factor of $1 + r$, meaning "the new value is $(1 + r)$ times the old value"):

Year	Rate	Growth factor ($1 + r$)
1989	0.11	1.11
1990	0.09	1.09
1991	0.075	1.075
1992	0.08	1.08
1993	0.095	1.095
1994	0.108	1.108
1995	0.120	1.12

Step 2 — multiply all seven growth factors together:

$$1.11 \times 1.09 \times 1.075 \times 1.08 \times 1.095 \times 1.108 \times 1.12$$

Building this up term by term:

$$1.11 \times 1.09 = 1.2099$$

$$1.2099 \times 1.075 = 1.300643$$

$$1.300643 \times 1.08 = 1.404694$$

$$1.404694 \times 1.095 = 1.538140$$

$$1.538140 \times 1.108 = 1.704260$$

$$1.704260 \times 1.12 = 1.908771$$

So the product of all seven growth factors is approximately **1.908771**.

Step 3 — take the 7th root (there are 7 growth factors being averaged, so $n = 7$):

$$GM = \sqrt[7]{1.908771} = 1.09675$$

Final Answer (part 1): $GM \approx 1.09675$, meaning the *average* annual growth rate over 1989–1995 was about **9.675% per year**.

Step 4 — project forward to 1997. The data runs through 1995; 1997 is **2 years** later (1996 and 1997). If bad debt expense continues growing at the average rate found above, the total growth factor over those 2 years is the GM raised to the power of 2:

$$GM^2 = 1.09675^2$$

Computing this:

$$1.09675 \times 1.09675 = 1.202861$$

So the expense is projected to be about **1.2029 times** its 1995 level — i.e., it's expected to be **20.29% higher** than in 1995.

Final Answer (part 2): 20.29% higher than in 1995. Both parts match the given answer key exactly.

Question 6 — Average Monthly Rate of Retail Price Decrease

Given: The shop marks retail price up 35% above cost. Product A's cost had been \$300 (4 months ago); over the last 4 months, four shipments arrived at unit costs of \$275, \$250, \$240, \$225.

Setting up the problem: We're asked for the *retail price's* average rate of decrease, not the cost's — so the first step is to convert every cost figure into its corresponding retail price using the 35% markup rule:

$$\text{Retail price} = \text{cost} \times 1.35$$

Point in time	Unit Cost	Retail Price = cost $\times$ 1.35
4 months ago (baseline)	\$300	\$405.00
End of month 1	\$275	\$371.25
End of month 2	\$250	\$337.50
End of month 3	\$240	\$324.00

Point in time	Unit Cost	Retail Price = cost $\times$ 1.35
End of month 4	\$225	\$303.75

This gives us 5 retail-price data points spanning **4 monthly intervals** — exactly analogous to Question 5's setup, except here we're measuring *decline* rather than growth, so each month-over-month ratio will come out less than 1.

Step 1 — compute each month's ratio (new retail price $\div$ previous retail price):

$$\frac{371.25}{405.00} = 0.91667$$

$$\frac{337.50}{371.25} = 0.90909$$

$$\frac{324.00}{337.50} = 0.96000$$

$$\frac{303.75}{324.00} = 0.93750$$

Step 2 — multiply these four ratios together. Notice that consecutive terms telescope (each denominator cancels the previous numerator), so the product of all four ratios must equal simply the final retail price divided by the starting retail price:

$$0.91667 \times 0.90909 \times 0.96000 \times 0.93750 = \frac{303.75}{405.00} = 0.75$$

Step 3 — take the 4th root (4 monthly intervals, so $n = 4$):

$$GM = \sqrt[4]{0.75}$$

Computing this: $\ln(0.75) = -0.287682$; dividing by 4 gives -0.071921 ; and $e^{-0.071921} = 0.93061$.

$$GM \approx 0.9306$$

Final Answer (part 1): $GM \approx 0.9306$ — this matches the given answer.

Step 4 — convert to a percentage rate of decrease. A growth factor of 0.9306 means the price is, on average, only 93.06% of what it was the month before — i.e., it's falling by:

$$1 - 0.9306 = 0.0694 = 6.94\%$$

Final Answer (part 2): the retail price fell at an average rate of 6.94% per month. Both results match the given answer key.

Question 7 — Mode, Mean, and Comparison (Student Ages)

Given raw data (30 students' ages): 19, 17, 15, 20, 23, 41, 33, 21, 18, 20, 18, 33, 32, 29, 24, 19, 18, 20, 17, 22, 55, 19, 22, 25, 28, 30, 44, 19, 20, 39

Part (a) — Construct the frequency distribution

We need to sort all 30 ages into the five given intervals: 15–19, 20–24, 25–29, 30–34, and 35 and older. Going through the raw list one value at a time and placing a tally mark into the appropriate bucket:

- **15–19** (ages 15,16,17,18,19): 19, 17, 15, 18, 18, 19, 18, 17, 19, 19 → **10 students**
- **20–24** (ages 20,21,22,23,24): 20, 23, 21, 20, 24, 20, 22, 22, 20 → **9 students**
- **25–29** (ages 25,26,27,28,29): 29, 25, 28 → **3 students**
- **30–34** (ages 30,31,32,33,34): 33, 33, 32, 30 → **4 students**
- **35 and older**: 41, 55, 44, 39 → **4 students**

Check: $10 + 9 + 3 + 4 + 4 = 30$ ✓ — matches the total number of students, confirming no one was miscounted.

Age interval	Frequency
15–19	10
20–24	9
25–29	3
30–34	4
35 and older	4

Part (b) — Estimate the mode

Identifying the modal class: the class with the highest frequency is **15–19**, with 10 students — this is our modal class.

Formula:

$$Mode = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

Identifying each variable:

- $L = 15$ — the lower limit of the modal class (15–19). No half-unit continuity correction is applied here since the interval boundaries are used exactly as given.
- $f_1 = 10$ — the frequency of the modal class itself.

- $f_0 = 0$ — the frequency of the class *before* the modal class. Since 15–19 is the very first interval in the table, there is no preceding class, so this is treated as 0.
- $f_2 = 9$ — the frequency of the class *after* the modal class, which is 20–24.
- $h = 5$ — the width of each class interval (e.g., 15–19 spans 5 integer values, and more generally the interval width used in the formula is 5).

Plugging in:

$$Mode = 15 + \left(\frac{10 - 0}{2(10) - 0 - 9} \right) \times 5 = 15 + \left(\frac{10}{20 - 9} \right) \times 5 = 15 + \left(\frac{10}{11} \right) \times 5$$

$$Mode = 15 + \frac{50}{11} = 15 + 4.5454... = 19.545...$$

Final Answer: Mode $\approx$ 19.55 years — matches the given answer.

Part (c) — Compute the mean of the raw data

Since we have all 30 individual raw values (not just grouped frequencies), we compute the simple arithmetic mean directly:

$$\bar{x} = \frac{\text{sum of all 30 ages}}{30}$$

Summing the raw data (breaking into the three rows of 10 as given):

$$\text{Row 1: } 19 + 17 + 15 + 20 + 23 + 41 + 33 + 21 + 18 + 20 = 227$$

$$\text{Row 2: } 18 + 33 + 32 + 29 + 24 + 19 + 18 + 20 + 17 + 22 = 232$$

$$\text{Row 3: } 55 + 19 + 22 + 25 + 28 + 30 + 44 + 19 + 20 + 39 = 301$$

Total sum:

$$227 + 232 + 301 = 760$$

Mean:

$$\bar{x} = \frac{760}{30} = 25.333...$$

Final Answer: Mean $\approx$ 25.33 years — matches the given answer.

Part (d) — Compare and comment

The mode (19.55) and the mean (25.33) are quite far apart — a gap of nearly 6 years. This gap is not an error; it's a direct signature of **skewness** in the data. Looking back at the raw list, most students cluster tightly in their late teens and early twenties (that's exactly why the 15–19 and 20–24 classes together hold 19 of the 30 students), but there are a handful of much older

outliers — ages 41, 44, 55, and 39 — sitting far above the rest of the group. Because the mean is computed by summing *every* value and dividing, those four large outliers pull the arithmetic mean upward substantially, away from where the bulk of the class actually sits. The mode, by contrast, is driven purely by where observations are most densely packed, and is completely untouched by how extreme the outlier values happen to be.

Conclusion: for this dataset, the **mode (19.55) is the better measure of central tendency** — it accurately reflects the age of a "typical" student in this class, whereas the mean (25.33) is distorted upward by a small number of older, non-representative students and doesn't describe any age group that most students actually belong to. (In general, for skewed distributions with outliers like this one, the median is also a strong choice for the same reason — it likewise resists being pulled around by extreme values — but between the two measures actually computed here, mode wins out as the more representative "typical value.")

Question 8 — Median from Cumulative Frequency Table (Age Distribution)

Given (cumulative "less than" frequencies, in thousands):

Age (years)	No. of persons (thousands)
Below 10	2
Below 20	5
Below 30	9
Below 40	12
Below 50	14
Below 60	15
Below 70	15.5
70 and over	15.6

Step 1 — recognize this as a "less than" cumulative frequency table, and recover the true total. The final row, "70 and over," represents the grand total population once all ages are accounted for: = 15.6 thousand persons (there is no additional data beyond this row, so this is the complete population).

Step 2 — locate the median class. The median is the value that splits the population exactly in half, so we first need:

$$\frac{\bar{x}}{2} = \frac{15.6}{2} = 7.8$$

We need to find which age interval contains the point where the cumulative count first reaches 7.8. Scanning down the cumulative column: "Below 20" = 5 (not yet 7.8), "Below 30" = 9 (this has passed 7.8). So the median falls somewhere in the interval **20–30**.

Step 3 — identify each variable for the median formula:

- $L = 20$ — the lower boundary of the median class (20–30).
- $\bar{x} = 15.6$ — total population (in thousands).
- $CF = 5$ — the cumulative frequency of everything *before* the median class (i.e., "Below 20").
- $f = 9 - 5 = 4$ — the frequency *within* the median class alone (not cumulative): this is the count "Below 30" minus the count "Below 20," representing just the people aged 20–30.
- $h = 10$ — the width of the median class (20 to 30).

Formula:

$$Median = L + \left(\frac{\bar{x} - CF}{f} \right) \times h$$

Plugging in:

$$Median = 20 + \left(\frac{7.8 - 5}{4} \right) \times 10 = 20 + \left(\frac{2.8}{4} \right) \times 10 = 20 + 0.7 \times 10 = 20 + 7 = 27$$

Final Answer: Median = 27 years — matches the given answer.

Part (ii) — why is the median more suitable than the mean here? The very last class, "70 and over," is **open-ended** — it has no stated upper boundary. To compute a mean, you would need to know (or assume) a representative midpoint for every class, but an open-ended class has no natural midpoint (is it centered at 75? 80? 90?), so any mean computed from this table would depend heavily on an arbitrary assumption about the oldest group. The median sidesteps this problem entirely: it only requires locating *which* class contains the middle observation and doing simple interpolation within that class — it never needs to know anything about the open-ended tail beyond how many people fall into it. This makes the median far more reliable and less sensitive to arbitrary assumptions whenever a distribution has an open-ended class.

Question 9 — Finding a Missing Frequency Given the Median (Shop Sales)

Given (sale figures in hundreds of Rs.):

Sale (hundred Rs.)	0–10	10–20	20–30	30–40	40–50
No. of shops	5	25	(missing, call it f)	18	7

Median sale = Rs. 2,400.

Step 1 — convert the median into the same units as the table. The table's "sale" axis is measured in *hundreds of rupees*, so Rs. 2,400 corresponds to $2400/100 = 24$ on this table's scale.

Step 2 — locate the median class. A value of 24 falls between the class boundaries 20 and 30, so the median class is **20–30** — which is conveniently exactly the class with the missing frequency, so finding f is really the whole point of this problem.

Step 3 — set up the total. Let be the (currently unknown) total number of shops:

$$= 5 + 25 + f + 18 + 7 = 55 + f$$

Step 4 — identify the median-formula variables in terms of f :

- $L = 20$ — lower boundary of the median class.
- $CF = 5 + 25 = 30$ — cumulative frequency of everything before the median class.
- f — frequency of the median class itself (this is our unknown).
- $h = 10$ — class width.
- $\frac{55+f}{2}$ — half the (also unknown) total.

Step 5 — plug everything into the median formula and set it equal to 24:

$$24 = 20 + \left(\frac{\frac{55+f}{2} - 30}{f} \right) \times 10$$

Step 6 — solve algebraically. Subtract 20 from both sides:

$$4 = \left(\frac{\frac{55+f}{2} - 30}{f} \right) \times 10$$

Divide both sides by 10:

$$0.4 = \frac{\frac{55+f}{2} - 30}{f}$$

Multiply both sides by f :

$$0.4f = \frac{55+f}{2} - 30$$

Multiply everything by 2 to clear the fraction:

$$0.8f = 55 + f - 60$$

$$0.8f = f - 5$$

Subtract f from both sides:

$$-0.2f = -5$$

Divide by -0.2 :

$$f = 25$$

Final Answer: the missing frequency is 25 shops — matches the given answer exactly.

Verification: with $f = 25$, total shops = $55 + 25 = 80$, so $/2 = 40$. The median class 20–30 spans cumulative frequency from $CF = 30$ up to $30 + 25 = 55$, and 40 does fall inside that range — confirming 20–30 really is the correct median class, and the algebra above is internally consistent.

Question 10 — Finding Two Missing Frequencies Given the Median (Family Expenditure)

Given (100 families total; median expenditure = 50):

Expenditure	0–20	20–40	40–60	60–80	80–100
No. of families	14	f_1 (missing)	27	f_2 (missing)	15

Step 1 — use the total to relate f_1 and f_2 . Since all frequencies must sum to the given total of 100 families:

$$14 + f_1 + 27 + f_2 + 15 = 100$$

$$f_1 + f_2 = 100 - 14 - 27 - 15 = 44$$

This is our first equation, linking the two unknowns. We need a second equation — which comes from the median condition — to pin down each one individually.

Step 2 — locate the median class. The median value, 50, falls between 40 and 60, so the median class is **40–60** — and conveniently, this class's frequency (27) is already known, which is what lets us solve for f_1 using the median formula.

Step 3 — identify the median-formula variables:

- $L = 40$ — lower boundary of the median class.
- $n = 100$, so $\frac{n}{2} = 50$.
- $CF = 14 + f_1$ — cumulative frequency of everything before the median class (0–20 plus 20–40).
- $f = 27$ — frequency of the median class itself.
- $h = 20$ — class width.

Step 4 — plug into the median formula and set equal to 50:

$$50 = 40 + \left(\frac{50 - (14 + f_1)}{27} \right) \times 20$$

Step 5 — solve algebraically. Subtract 40 from both sides:

$$10 = \left(\frac{50 - 14 - f_1}{27} \right) \times 20 = \left(\frac{36 - f_1}{27} \right) \times 20$$

Divide both sides by 20:

$$0.5 = \frac{36 - f_1}{27}$$

Multiply both sides by 27:

$$\begin{aligned} 13.5 &= 36 - f_1 \\ f_1 &= 36 - 13.5 = 22.5 \end{aligned}$$

Step 6 — round to a whole number. Since f_1 represents a count of families, it must be a whole number — the algebra produces 22.5 because the underlying median formula is a linear-interpolation *approximation* (it assumes expenditure is spread perfectly evenly within the median class), and real integer family counts don't always land exactly on that idealized line. Rounding to the nearest whole family:

$$f_1 = 23$$

Step 7 — solve for f_2 using the total-frequency equation from Step 1:

$$f_2 = 44 - f_1 = 44 - 23 = 21$$

Final Answer: $f_1 = 23$, $f_2 = 21$ — matches the given answer.

Verification: $14 + 23 + 27 + 21 + 15 = 100$ ✓ (total families check out).

ASSIGNMENT 2 — Unit 4: Probability

Question 1 — Four Cards Drawn from a Standard Deck

Setup common to all six parts: we're drawing 4 cards at random from a standard 52-card deck, without replacement, and order doesn't matter. The total number of equally likely ways to choose any 4 cards out of 52 is:

$$\binom{52}{4} = \frac{52!}{4!(52-4)!} = \frac{52 \times 51 \times 50 \times 49}{4 \times 3 \times 2 \times 1} = 270,725$$

This number — **270,725** — is the denominator for every part of this question, since it represents the full sample space of possible 4-card hands.

(i) One king, one queen, one jack, and one ace

We need exactly one card of each of these four specific ranks. There are 4 kings in the deck, and we need to choose exactly 1 of them: $\binom{4}{1} = 4$ ways. The same logic applies independently to queens, jacks, and aces — each also has $\binom{4}{1} = 4$ ways to pick one.

Since these four choices are made independently of each other (choosing which king doesn't affect which queen you can choose), the total number of favorable hands is found by **multiplying** the number of choices for each rank:

$$4 \times 4 \times 4 \times 4 = 256$$
$$P(\text{one of each: ,,,}) = \frac{256}{270,725} \quad 0.000946 \text{ (} 0.0946$$

(ii) Two kings and two aces

We need exactly 2 of the 4 kings, and exactly 2 of the 4 aces (and nothing else). The number of ways to choose 2 kings out of 4 is $\binom{4}{2} = 6$; likewise for aces, $\binom{4}{2} = 6$. Multiplying these independent choices:

$$6 \times 6 = 36$$
$$P(2 \text{ kings and } 2 \text{ aces}) = \frac{36}{270,725} \quad 0.000133 \text{ (} 0.0133$$

(iii) All four cards are diamonds

There are 13 diamonds in the deck, and we need to choose all 4 cards from just those 13:

$$\binom{13}{4} = \frac{13 \times 12 \times 11 \times 10}{4 \times 3 \times 2 \times 1} = 715$$
$$P(\text{all diamonds}) = \frac{715}{270,725} \quad 0.002641 \text{ (} 0.2641$$

(iv) Two red and two black

There are 26 red cards (hearts + diamonds) and 26 black cards (clubs + spades). We need 2 of the 26 red cards and 2 of the 26 black cards:

$$\binom{26}{2} = \frac{26 \times 25}{2} = 325 \quad (\text{for red}), \quad \binom{26}{2} = 325 \quad (\text{for black})$$

Multiplying these two independent selections:

$$325 \times 325 = 105,625$$

$$P(2 \text{ red, 2 black}) = \frac{105,625}{270,725} \approx 0.3902 \quad (39.02\%)$$

(v) One card of each suit

There are 4 suits (hearts, diamonds, clubs, spades), each with 13 cards. We need exactly one card from each suit — choosing 1 of 13 hearts, 1 of 13 diamonds, 1 of 13 clubs, and 1 of 13 spades, independently:

$$13 \times 13 \times 13 \times 13 = 13^4 = 28,561$$

$$P(\text{one of each suit}) = \frac{28,561}{270,725} \approx 0.1055 \quad (10.55\%)$$

(vi) Two clubs and two diamonds

We need 2 of the 13 clubs and 2 of the 13 diamonds:

$$\binom{13}{2} = \frac{13 \times 12}{2} = 78 \quad (\text{for clubs}), \quad \binom{13}{2} = 78 \quad (\text{for diamonds})$$

Multiplying:

$$78 \times 78 = 6,084$$

$$P(2 \text{ clubs, 2 diamonds}) = \frac{6,084}{270,725} \approx 0.02248 \quad (2.248\%)$$

Question 2 — Ticket Drawn from a Bag of 20

Given: 20 tickets numbered 1 to 20; one is drawn at random.

(i) Probability the number is a multiple of 2 or 5

Multiples of 2 among 1–20: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 → **10 numbers**, so

$$P(\text{multiple of 2}) = 10/20.$$

Multiples of 5 among 1–20: 5, 10, 15, 20 → **4 numbers**, so $P(\text{multiple of 5}) = 4/20$.

Overlap (multiples of BOTH 2 and 5, i.e., multiples of 10): 10, 20 → **2 numbers**, so

$P(\text{multiple of 2 and 5}) = 2/20$.

We can't just add $10/20 + 4/20$ directly, because numbers 10 and 20 would then be counted *twice* (they're multiples of both 2 and 5). The **addition rule** corrects for this by subtracting the overlap exactly once:

$$P(2 \text{ or } 5) = P(2) + P(5) - P(2 \text{ and } 5) = \frac{10}{20} + \frac{4}{20} - \frac{2}{20} = \frac{12}{20} = 0.6$$

Final Answer: 0.6 — matches given.

(ii) Probability the number is a multiple of 3 or 5

Multiples of 3 among 1–20: 3, 6, 9, 12, 15, 18 → **6 numbers**.

Multiples of 5 among 1–20: 5, 10, 15, 20 → **4 numbers** (as before).

Overlap (multiples of both 3 and 5, i.e., multiples of 15): 15 → **1 number**.

$$P(3 \text{ or } 5) = \frac{6}{20} + \frac{4}{20} - \frac{1}{20} = \frac{9}{20} = 0.45$$

Final Answer: 0.45 — matches given.

Question 3 — Joint Probability Table (Manufacturers X and Y)

Given joint probability table:

Manufacturer	0 defects	1 defect	2 defects	3 defects
X	0.1250	0.0625	0.1875	0.1250
Y	0.0625	0.0625	0.1250	0.2500

Sanity check first: all eight cells should sum to exactly 1 (since this table covers every possible outcome — every part comes from either X or Y, and has 0, 1, 2, or 3 defects).

Summing: $0.125 + 0.0625 + 0.1875 + 0.125 + 0.0625 + 0.0625 + 0.125 + 0.25 = 1.0$ ✓. Good — this confirms the table is internally consistent before we compute anything from it.

(i) $P(X \mid 0 \text{ defects})$

We're told a part has 0 defects, and asked: given that, what's the probability it came from manufacturer X? This is a conditional probability:

$$P(X | 0 \text{ defects}) = \frac{P(X \cap 0 \text{ defects})}{P(0 \text{ defects})}$$

Numerator — read directly off the table, the joint probability of (X, 0 defects):

$$P(X \cap 0) = 0.1250$$

Denominator — the *marginal* probability of 0 defects, found by adding up the 0-defects column across both manufacturers:

$$P(0 \text{ defects}) = P(X \cap 0) + P(Y \cap 0) = 0.1250 + 0.0625 = 0.1875$$

Putting it together:

$$P(X | 0) = \frac{0.1250}{0.1875} = 0.6667$$

Final Answer: 0.6667 — matches given.

(ii) P(0 defects | X)

This is the reverse conditional — given the part came from manufacturer X, what's the chance it has no defects?

$$P(0 | X) = \frac{P(X \cap 0)}{P(X)}$$

Numerator: $P(X \cap 0) = 0.1250$ (same joint cell as before).

Denominator — the marginal probability of manufacturer X, found by adding up X's entire row:

$$P(X) = 0.1250 + 0.0625 + 0.1875 + 0.1250 = 0.5000$$

Putting it together:

$$P(0 | X) = \frac{0.1250}{0.5000} = 0.25$$

Final Answer: 0.25 — matches given. (Note this is a genuinely different question from part (i) — conditional probability is *not* symmetric; "probability X given 0 defects" and "probability 0 defects given X" are two entirely different numbers, as we've just seen: 0.6667 vs. 0.25.)

(iii) P(X | 2 or more defects)

Numerator — the joint probability of (X, 2 or more defects), found by adding X's 2-defect and 3-defect cells:

$$P(X \cap 2) = 0.1875 + 0.1250 = 0.3125$$

Denominator — the marginal probability of 2 or more defects overall (both manufacturers combined):

$$P(2) = P(\cap 2) + P(\cap 3) + P(\cap 2) + P(\cap 3) = 0.1875 + 0.1250 + 0.1250 + 0.2500 = 0.6875$$

Putting it together:

$$P(2) = \frac{0.3125}{0.6875} = 0.4545$$

Final Answer: 0.4545 — matches given.

(iv) $P(Y \mid 1 \text{ or more defects})$

Numerator — the joint probability of (Y, 1 or more defects), found by adding Y's 1-, 2-, and 3-defect cells:

$$P(\cap 1) = 0.0625 + 0.1250 + 0.2500 = 0.4375$$

Denominator — the marginal probability of 1 or more defects overall. The easiest way to get this is to note that "1 or more defects" is the complement of "0 defects," so:

$$P(1) = 1 - P(0) = 1 - 0.1875 = 0.8125$$

Putting it together:

$$P(1) = \frac{0.4375}{0.8125} = 0.5385$$

Final Answer: 0.5385 — matches given.

Question 4 — Filling In a Probability Table (Economists' Opinions)

Given (partial) table:

Economists	Stable (S)	Expanding (E)	Contracting (C)	Total
Academics (A)	25	?	20	?
Private Sector (R)	?	7	?	22
Government (G)	5	8	?	13
Total	40	?	?	100

Step 1 — fill in the Private Sector row. We know the row total is 22 and the Expanding cell is 7, so:

$$R + 7 + C_R = 22 \quad R + C_R = 15$$

(this gives us one equation relating the two unknowns in this row — we'll need the column total to pin down R exactly).

Step 2 — fill in the Government row. We know the row total is 13, Stable is 5, and Expanding is 8:

$$5 + 8 + C_G = 13 \quad C_G = 0$$

So the Government row is now fully known: 5, 8, 0, total 13.

Step 3 — use the Stable column total to solve for R . We're told the Stable column sums to 40:

$$25 + R + 5 = 40 \quad R = 40 - 25 - 5 = 10$$

Step 4 — go back to the Private Sector row equation from Step 1 to now find C_R :

$$R + C_R = 15 \quad 10 + C_R = 15 \quad C_R = 5$$

The Private Sector row is now fully known: 10, 7, 5, total 22.

Step 5 — use the grand total to solve for the Academics row total. All row totals must sum to the grand total of 100:

$$A + 22 + 13 = 100 \quad A = 100 - 22 - 13 = 65$$

Step 6 — use the Academics row total to solve for A :

$$25 + A + 20 = 65 \quad A = 65 - 45 = 20$$

The Academics row is now fully known: 25, 20, 20, total 65.

Step 7 — fill in the remaining column totals by summing down each column:

$$\text{Expanding total} = 20 + 7 + 8 = 35$$

$$\text{Contracting total} = 20 + 5 + 0 = 25$$

Check: $40 + 35 + 25 = 100$ ✓ matches the grand total.

Fully completed table:

Economists	Stable	Expanding	Contracting	Total
Academics (A)	25	20	20	65
Private Sector (R)	10	7	5	22

Economists	Stable	Expanding	Contracting	Total
Government (G)	5	8	0	13
Total	40	35	25	100

Now every requested probability is a simple lookup (dividing counts by the grand total of 100 economists to convert them into probabilities):

(i) **P(A)**: Academics total ÷ grand total = $65/100 = .$

(ii) **P(E)**: Expanding total ÷ grand total = $35/100 = .$

(iii) **P(A ∩ E)**: the joint (Academics AND Expanding) cell ÷ grand total = $20/100 = .$

(iv) **P(G ∩ C)**: the joint (Government AND Contracting) cell ÷ grand total = $0/100 =$

(v) **P(E | G)**: conditional — given Government, probability of Expanding:

$$P(E | G) = \frac{P(G \cap E)}{P(G)} = \frac{8/100}{13/100} = \frac{8}{13} = .$$

(vi) **P(G | E)**: the reverse conditional — given Expanding, probability of Government:

$$P(G | E) = \frac{P(G \cap E)}{P(E)} = \frac{8/100}{35/100} = \frac{8}{35} = .$$

All six results match the given answer key (0.65, 0.35, 0.20, 0, 0.6154, 0.2286).

Question 5 — Household Survey (TV Ownership & Telephone Subscription)

Given (125 households total), split by income level:

Income ≤ Rs. 8,000:

	Telephone Subscriber	No Telephone
Own TV set	27	20
No TV set	18	10

Income > Rs. 8,000:

	Telephone Subscriber	No Telephone
Own TV set	18	10
No TV set	12	10

Step 1 — combine both income groups into one overall table, since several of the questions ask about the household population as a whole, not split by income:

$$\text{Own TV \& Telephone} = 27 + 18 = 45$$

$$\text{Own TV \& No Telephone} = 20 + 10 = 30$$

$$\text{No TV \& Telephone} = 18 + 12 = 30$$

$$\text{No TV \& No Telephone} = 10 + 10 = 20$$

Combined table:

	Telephone Subscriber	No Telephone	Row Total
Own TV set	45	30	75
No TV set	30	20	50
Column Total	75	50	125

Check: $45 + 30 + 30 + 20 = 125$ ✓ matches the given total number of households.

(i) Probability of drawing a TV owner at random

This is simply the marginal probability of TV ownership, from the combined table:

$$P(T) = \frac{75}{125} = .$$

(ii) Given income > Rs. 8,000 AND a telephone subscriber, probability of owning a TV

This question restricts us to only the *income > 8,000* sub-table (not the combined one), and within that group, further restricts to telephone subscribers only:

From the income > 8,000 table: Telephone subscribers with TV = 18; Telephone subscribers without TV = 12. Total telephone subscribers in this income bracket:

$$18 + 12 = 30$$

$$P(T \mid \text{income} > 8000, \text{telephone}) = \frac{18}{30} = .$$

(iii) Conditional probability of owning a TV, given the household is a telephone subscriber (overall, any income)

This uses the **combined** table, since income level isn't mentioned in this part:

$$P(T \mid \text{Telephone}) = \frac{P(T \cap \text{Telephone})}{P(\text{Telephone})} = \frac{45/125}{75/125} = \frac{45}{75} = .6$$

(iv) Are "TV ownership" and "telephone subscription" statistically independent?

Recall two events are independent exactly when $P(A \cap B) = P(A) \times P(B)$ (equivalently, when the conditional probability equals the plain/marginal probability). We already have both pieces we need:

$$P(T) = 0.6 \quad (\text{from art i})$$

$$P(T \mid \text{Telephone}) = 0.6 \quad (\text{from art iii})$$

These are **exactly equal** — knowing that a household is a telephone subscriber tells you nothing extra about whether it owns a TV, beyond what you'd already expect from the population as a whole. We can double check this with the direct independence formula too:

$$P(T) \times P(\text{Telephone}) = 0.6 \times \frac{75}{125} = 0.6 \times 0.6 = 0.36$$

$$P(T \cap \text{Telephone}) = \frac{45}{125} = 0.36$$

These match exactly. **Conclusion: yes, based on this data, TV ownership and telephone subscription are statistically independent events** — the two "habits" don't influence one another in this sample.

Question 6 — Golf Gloves (Left- vs. Right-Handed)

Given: a box of 9 gloves: 2 left-handed, 7 right-handed.

(i)(a) Both gloves right-handed, drawn without replacement

For the first draw, 7 of the 9 gloves are right-handed, so $P(\text{1st right}) = 7/9$. Once that glove is removed (without replacement), only 8 gloves remain in the box, of which 6 are right-handed, so $P(\text{2nd right} \mid \text{1st right}) = 6/8$.

Since we need *both* events to happen in sequence, we multiply:

$$P(\text{both right}) = \frac{7}{9} \times \frac{6}{8} = \frac{42}{72} = .$$

(i)(b) One left-handed and one right-handed (in either order)

There are two distinct ways this can happen — left-then-right, or right-then-left — and since these two scenarios are mutually exclusive, we add their probabilities.

Left then right: $P(\text{1st left}) = 2/9$; after removing a left glove, 8 remain (7 right, 1 left), so $P(\text{2nd right} \mid \text{1st left}) = 7/8$.

$$P(\text{ then R}) = \frac{2}{9} \times \frac{7}{8} = \frac{14}{72}$$

Right then left: $P(\text{1st right}) = 7/9$; after removing a right glove, 8 remain (6 right, 2 left), so $P(\text{2nd left} \mid \text{1st right}) = 2/8$.

$$P(\text{R then }) = \frac{7}{9} \times \frac{2}{8} = \frac{14}{72}$$

Adding both orders together:

$$P(\text{one of each}) = \frac{14}{72} + \frac{14}{72} = \frac{28}{72} = .$$

(ii) Three gloves selected without replacement — all left-handed

There are only **2** left-handed gloves in the entire box. Since we're selecting 3 gloves and requiring *all three* to be left-handed, but there physically aren't 3 left-handed gloves available to draw, this event is **impossible**.

$$P(\text{all 3 lefthanded}) =$$

(iii) Two gloves selected WITH replacement — both right-handed

With replacement means the glove is put back (and the box "reset") after each draw, so both draws are identical, independent trials, each with $P(\text{right}) = 7/9$:

$$P(\text{both right, ith relacement}) = \frac{7}{9} \times \frac{7}{9} = \frac{49}{81} = .$$

Question 7 — Drawing Balls from One of Two Bags

Given: Bag 1 has 5 white + 3 black = 8 balls total. Bag 2 has 4 white + 5 black = 9 balls total. A bag is chosen at random (so each bag has a 1/2 chance of being picked), then two balls are

drawn together from that chosen bag.

Why this needs the law of total probability: the final answer depends on *which* bag gets picked, and the two bags have different compositions — so we compute the probability of "one white, one black" separately for each bag, then combine the two results weighted by how likely each bag is to be chosen (1/2 each).

Step 1 — probability of one white and one black ball, IF Bag 1 is used.

The number of ways to draw any 2 balls out of Bag 1's 8 total is $\binom{8}{2} = \frac{8 \times 7}{2} = 28$.

The number of ways to get exactly one white AND one black is: choose 1 of the 5 white balls, and 1 of the 3 black balls:

$$\binom{5}{1} \times \binom{3}{1} = 5 \times 3 = 15$$

$$P(1 \text{ hite}, 1 \text{ black} \mid \text{ag 1}) = \frac{15}{28} = 0.5357$$

Step 2 — probability of one white and one black ball, IF Bag 2 is used.

The number of ways to draw any 2 balls out of Bag 2's 9 total is $\binom{9}{2} = \frac{9 \times 8}{2} = 36$.

The number of ways to get exactly one white AND one black is: choose 1 of the 4 white balls, and 1 of the 5 black balls:

$$\binom{4}{1} \times \binom{5}{1} = 4 \times 5 = 20$$

$$P(1 \text{ hite}, 1 \text{ black} \mid \text{ag 2}) = \frac{20}{36} = 0.5556$$

Step 3 — combine using the law of total probability. Since each bag is equally likely to be selected (probability 1/2 each), and these are the only two possible "paths" to the final outcome:

$$\begin{aligned} P(1 \text{ hite}, 1 \text{ black}) &= P(\text{ag 1}) \times P(11b \mid \text{ag1}) + P(\text{ag 2}) \times P(11b \mid \text{ag2}) \\ &= \left(\frac{1}{2} \times 0.5357 \right) + \left(\frac{1}{2} \times 0.5556 \right) = 0.2679 + 0.2778 = . \end{aligned}$$

Final Answer: 0.5456 — matches the given answer.

ASSIGNMENT 3 — Unit 8: Bayes' Theorem

Question 1 — Advertising Campaign (TV, Radio, Print)

Given: 30% of the area watches TV ads, 50% listens to radio ads, and the remaining 20% relies on newspapers (print). Given exposure via each channel, the probability of buying is: 0.6 (TV), 0.3 (radio), 0.1 (print). A random consumer is found to have purchased the product. What's the probability she heard about it on the radio?

Why this needs Bayes' theorem: we're given probabilities in the "forward" direction (probability of *buying*, given a particular exposure channel), but we're asked for the "backward" direction — given that she *did* buy, what's the probability she was exposed via radio specifically? This reversal — going from "effect given cause" to "cause given effect" — is exactly what Bayes' theorem handles.

Step 1 — identify priors (probability of each exposure channel):

$$P(T) = 0.30, \quad P(\text{Radio}) = 0.50, \quad P(\text{Print}) = 0.20$$

Step 2 — identify the likelihoods (probability of buying, given each channel):

$$P(\text{uy} | T) = 0.6, \quad P(\text{uy} | \text{Radio}) = 0.3, \quad P(\text{uy} | \text{Print}) = 0.1$$

Step 3 — compute the overall probability of buying by combining every channel's contribution (each channel's share of the population, times its own buy-probability):

$$\begin{aligned} P(\text{uy}) &= P(T)P(\text{uy} | T) + P(\text{Radio})P(\text{uy} | \text{Radio}) + P(\text{Print})P(\text{uy} | \text{Print}) \\ &= (0.30 \times 0.6) + (0.50 \times 0.3) + (0.20 \times 0.1) = 0.18 + 0.15 + 0.02 = 0.35 \end{aligned}$$

Step 4 — apply Bayes' theorem to find the probability that a buyer specifically came from the radio channel:

$$P(\text{Radio} | \text{uy}) = \frac{P(\text{Radio}) \times P(\text{uy} | \text{Radio})}{P(\text{uy})} = \frac{0.50 \times 0.3}{0.35} = \frac{0.15}{0.35} = \frac{3}{7} \approx 0.4286$$

Final Answer: $\frac{3}{7}$ (≈ 0.4286 , or 42.86%) — matches the given answer.

Question 2 — Most Probable Cause of a Fire

Given: $P(\text{short circuit}) = 0.8$; $P(\text{LPG explosion}) = 0.2$. Given a short circuit, the chance of a fire is 30%; given an LPG explosion, the chance of a fire is 95%. A fire did occur — which cause is statistically more probable?

Why this is tricky (and why intuition can mislead you): LPG explosion has a *much* higher chance of causing a fire once it happens (95% vs. 30%) — so it's tempting to jump straight to "LPG explosion must be the more likely cause." But that reasoning ignores how *common* each

cause is in the first place. Short circuits happen far more often (80% base rate vs. just 20% for LPG explosions), so even though each individual short circuit is less likely to result in fire, there are so many more short circuits overall that they could still end up responsible for more fires in total. This is exactly the kind of situation Bayes' theorem is built to resolve correctly.

Step 1 — compute the overall probability of a fire occurring, combining both possible causes:

$$\begin{aligned} P(\text{ire}) &= P(\text{Short circuit}) \times P(\text{ire} \mid \text{Short circuit}) + P(\text{PG}) \times P(\text{ire} \mid \text{PG}) \\ &= (0.8 \times 0.3) + (0.2 \times 0.95) = 0.24 + 0.19 = 0.43 \end{aligned}$$

Step 2 — apply Bayes' theorem to find the posterior probability of each cause, given that a fire occurred:

$$\begin{aligned} P(\text{Short circuit} \mid \text{ire}) &= \frac{P(\text{Short circuit}) \times P(\text{ire} \mid \text{Short circuit})}{P(\text{ire})} = \frac{0.24}{0.43} = 0.5581 \\ P(\text{PG} \mid \text{ire}) &= \frac{P(\text{PG}) \times P(\text{ire} \mid \text{PG})}{P(\text{ire})} = \frac{0.19}{0.43} = 0.4419 \end{aligned}$$

Check: $0.5581 + 0.4419 = 1.0$ ✓ (these two posteriors must add up to 1, since short circuit and LPG explosion are treated as the only two possible causes here).

Conclusion: short circuit is the statistically more probable cause of the fire, at about 55.8% posterior probability, versus 44.2% for the LPG explosion — even though an LPG explosion is individually a "more dangerous" trigger (95% vs 30% chance of causing fire), its much lower prior frequency (only 20% of cases vs. 80% for short circuits) means it ends up explaining a smaller share of fires overall. This is a classic illustration of why the *base rate* of a cause matters just as much as how "effective" that cause is at producing the observed outcome.

Question 3 — The Two-Headed Coin

Given: 5 coins total — 1 has two heads (always lands heads), and 4 are normal/fair coins. One coin is picked at random and tossed 3 times.

(i) Probability of getting heads all three times

Step 1 — identify the two "paths" to this outcome and their prior probabilities: picking the two-headed coin ($P = 1/5$), or picking one of the four normal coins ($P = 4/5$).

Step 2 — find the likelihood of "HHH" under each path:

- If the two-headed coin was picked, it *always* lands heads, so $P(\text{ | toheaded}) = 1$.
- If a normal (fair) coin was picked, each toss independently has a $1/2$ chance of heads, so three heads in a row has probability $P(\text{ | normal}) = (1/2)^3 = 1/8 = 0.125$.

Step 3 — combine using the law of total probability:

$$\begin{aligned} P() &= P(\text{toheaded}) \times P(\text{ | toheaded}) + P(\text{normal}) \times P(\text{ | normal}) \\ &= \left(\frac{1}{5} \times 1 \right) + \left(\frac{4}{5} \times 0.125 \right) = 0.2 + 0.1 = .3 \end{aligned}$$

Final Answer: 0.3 — matches the given answer.

(ii) Given that it fell heads all three times, probability it's the two-headed coin

This is the reverse question — now that we've *observed* HHH, we update our belief about which coin was actually picked, using Bayes' theorem:

$$P(\text{toheaded | }) = \frac{P(\text{toheaded}) \times P(\text{ | toheaded})}{P()} = \frac{0.2 \times 1}{0.3} = \frac{0.2}{0.3} = .67$$

Final Answer: ≈ 0.67 — matches the given answer. Notice how much our belief shifted: before any tosses, there was only a 1-in-5 (20%) chance the two-headed coin was picked; after observing three heads in a row, that probability jumped to about two-thirds (67%) — a large but not certain update, because normal coins can still occasionally produce HHH by chance (with probability 0.125), so the evidence, while strong, isn't fully conclusive.

Question 4 — Three Jewelry Boxes (Bertrand's Box-Style Puzzle)

Given: three identical boxes, each with 2 drawers.

- **Box 1:** both drawers contain a gold watch (Gold-Gold)
- **Box 2:** both drawers contain a silver watch (Silver-Silver)
- **Box 3:** one drawer has gold, the other has silver (Gold-Silver)

A box is selected at random, one of its two drawers is opened at random, and it's found to contain a **silver** watch. What's the probability that the *other* drawer (still unopened) in that same box contains gold?

Why this is a classic "trick" question: a very common (but wrong) instinct is to say "well, it's either Box 2 (both silver) or Box 3 (mixed), so it's 50/50 whether the other drawer is gold." This

reasoning is flawed because it treats "Box 2" and "Box 3" as equally likely *given that we saw silver* — but Box 2 is actually more likely to produce a silver-drawer observation in the first place, since *both* of its drawers are silver (guaranteeing a silver observation), whereas Box 3 only has a silver drawer half the time. Bayes' theorem correctly accounts for this imbalance.

Step 1 — identify the prior probability of each box being chosen (all equally likely):

$$P(o\ 1) = P(o\ 2) = P(o\ 3) = \frac{1}{3}$$

Step 2 — identify the likelihood of observing a silver watch, given each box:

- **Box 1 (Gold-Gold):** neither drawer has silver, so $P(\text{silver} \mid o\ 1) = 0$.
- **Box 2 (Silver-Silver):** both drawers are silver, so no matter which drawer is opened, it's silver: $P(\text{silver} \mid o\ 2) = 1$.
- **Box 3 (Gold-Silver):** only one of the two drawers is silver, and the drawer is opened at random, so $P(\text{silver} \mid o\ 3) = 1/2$.

Step 3 — compute the overall probability of observing silver, combining all three boxes:

$$P(\text{silver}) = \frac{1}{3}(0) + \frac{1}{3}(1) + \frac{1}{3}\left(\frac{1}{2}\right) = 0 + \frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Step 4 — apply Bayes' theorem to find the probability that Box 3 (the mixed box) was chosen, given that silver was observed:

$$P(o\ 3 \mid \text{silver}) = \frac{P(o\ 3) \times P(\text{silver} \mid o\ 3)}{P(\text{silver})} = \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{2}} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{6} \times \frac{2}{1} = \frac{1}{3}$$

Step 5 — connect this back to the original question. If Box 3 (the mixed Gold-Silver box) is indeed the one that was chosen, then — since we already know the *opened* drawer was silver — the *other, unopened* drawer in that same box **must** contain gold (that's simply what "mixed box" means). So the probability that "the other drawer has gold" is exactly the same as the probability that "Box 3 was the box selected":

$$P(\text{other draer is gold} \mid \text{silver observed}) = P(o\ 3 \mid \text{silver}) = -$$

Final Answer: 1/3 — matches the given answer.

Why this is *not* 1/2, intuitively: observing silver is actually evidence that *slightly favors* Box 2 over Box 3 — because Box 2 was **guaranteed** to show silver (it has no gold drawer to possibly open instead), while Box 3 only had a 50-50 chance of showing silver rather than gold on this particular draw. This makes Box 2 more likely than Box 3 to be the explanation *after* seeing

silver — specifically twice as likely ($P(o\ 2 \mid \text{silver}) = 2/3$ vs. $P(o\ 3 \mid \text{silver}) = 1/3$), which is exactly why the final answer skews below the naive 50/50 guess.